

Creation Date 02-Feb-2010

Revision Date 29-Oct-2024

Revision Number 10

Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	<u>Cumyl hydroperoxide</u>
Cat No. :	349960000; 349960010; 349960050; 349962500
Synonyms	Cumene hydroperoxide
Index No	617-002-00-8
CAS No	80-15-9
EC No	201-254-7
Molecular Formula	C9 H12 O2

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

EU entity/business name
Thermo Fisher Scientific
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road,
Loughborough, Leicestershire LE11 5RG, United Kingdom

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

Section 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

ACR34996

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CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Organic peroxides

Type E (H242)

Health hazards

Aspiration Toxicity

Category 1 (H304)

Acute oral toxicity

Category 4 (H302)

Acute dermal toxicity

Category 4 (H312)

Acute Inhalation Toxicity - Vapors

Category 3 (H331)

Skin Corrosion/Irritation

Category 1 (H314) B

Serious Eye Damage/Eye Irritation

Category 1 (H318)

Carcinogenicity

Category 1B (H350)

Reproductive Toxicity

Category 1A (H360D)

Specific target organ toxicity - (repeated exposure)

Category 2 (H373)

Environmental hazards

Chronic aquatic toxicity

Category 2 (H411)

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H242 - Heating may cause a fire

H304 - May be fatal if swallowed and enters airways

H331 - Toxic if inhaled

H314 - Causes severe skin burns and eye damage

H350 - May cause cancer

H360D - May damage the unborn child

H373 - May cause damage to organs through prolonged or repeated exposure

H411 - Toxic to aquatic life with long lasting effects

H302 + H312 - Harmful if swallowed or in contact with skin

Combustible liquid

Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and

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easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician

Additional EU labelling
Restricted to professional users

2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors
Toxic to terrestrial vertebrates

Section 3: Composition/information on ingredients

3.1. Substances

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Cumene hydroperoxide	80-15-9	EEC No. 201-254-7	80-85	Org. Perox. E (H242) Acute Tox. 4 (H302) Acute Tox. 4 (H312) Acute Tox. 3 (H331) Skin Corr. 1B (H314) Eye Dam. 1 (H318) STOT RE 2 (H373) Aquatic Chronic 2 (H411)
Isopropylbenzene	98-82-8	EEC No. 202-704-5	7-13	Flam. Liq. 3 (H226) Asp. Tox. 1 (H304) STOT SE 3 (H335) Carc. 1B (H350) Aquatic Chronic 2 (H411)
Benzenemethanol, .alpha.,.alpha.-dimethyl-	617-94-7	EEC No. 210-539-5	5-8	Acute Tox. 4 (H302) Skin Irrit. 2 (H315) Eye Irrit. 2 (H319)
Acetophenone	98-86-2	EEC No. 202-708-7	0.5-1.5	Acute Tox. 4 (H302) Eye Irrit. 2 (H319)
Peroxide, bis(1-methyl-1-phenylethyl)	80-43-3	EEC No. 201-279-3	0.46-0.65	Org. Perox. F (H242) Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) Repr. 1A (H360D) Aquatic Chronic 2 (H411)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Cumene hydroperoxide	Eye Dam. 1 (H318) :: 3%<=C<10% Eye Irrit. 2 (H319) :: 1%<=C<3% Skin Corr. 1B (H314) :: C>=10% Skin Irrit. 2 (H315) :: 3%<=C<10% STOT SE 3 (H335) :: C<10%	-	-

Components	Reach Registration Number
Cumene hydroperoxide ;	01-2119475796-19

Full text of Hazard Statements: see section 16

Section 4: First aid measures

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4.1. Description of first aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.
Ingestion	Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.
Inhalation	If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Risk of serious damage to the lungs (by aspiration). Remove to fresh air. Immediate medical attention is required.
Self-Protection of the First Aider	Use personal protective equipment as required.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

Section 5: Firefighting measures

5.1. Extinguishing media

Suitable Extinguishing Media

Water mist may be used to cool closed containers. CO₂, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.). Combustible material. Containers may explode when heated.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂).

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

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Section 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Take precautionary measures against static discharges. Remove all sources of ignition.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal. Remove all sources of ignition.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

Section 7: Handling and storage

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe mist/vapors/spray. Keep away from clothing and other combustible materials. Keep away from open flames, hot surfaces and sources of ignition.

Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

7.2. Conditions for safe storage, including any incompatibilities

Keep refrigerated. Store in a well-ventilated place. Keep cool. Keep container tightly closed. Keep away from heat, sparks and flame. Do not store near combustible materials. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Storage Class/LGK 5.2

Switzerland - Storage of hazardous substances

Storage class - SC 5
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

Section 8: Exposure controls/personal protection

8.1. Control parameters

Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work

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(Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Isopropylbenzene		STEL: 50 ppm 15 min STEL: 250 mg/m ³ 15 min TWA: 25 ppm 8 hr TWA: 125 mg/m ³ 8 hr Skin	TWA / VME: 10 ppm (8 heures). restrictive limit TWA / VME: 50 mg/m ³ (8 heures). restrictive limit TWA / VME: 150 mg/m ³ (8 heures). TWA / VME: 1000 mg/m ³ (8 heures). STEL / VLCT: 50 ppm. restrictive limit: If biological testing is set up, exposure is monitored using the biological test values available and appropriate for the chemical agent STEL / VLCT: 250 mg/m ³ . restrictive limit: If biological testing is set up, exposure is monitored using the biological test values available and appropriate for the chemical agent STEL / VLCT: 1500 mg/m ³ . Peau	TWA: 10 ppm 8 uren TWA: 50 mg/m ³ 8 uren STEL: 50 ppm 15 minuten STEL: 250 mg/m ³ 15 minuten Huid	STEL / VLA-EC: 50 ppm (15 minutos). STEL / VLA-EC: 250 mg/m ³ (15 minutos). TWA / VLA-ED: 10 ppm (8 horas) TWA / VLA-ED: 50 mg/m ³ (8 horas) Piel
Acetophenone				TWA: 10 ppm 8 uren TWA: 50 mg/m ³ 8 uren	TWA / VLA-ED: 10 ppm (8 horas) TWA / VLA-ED: 50 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Isopropylbenzene	TWA: 10 ppm 8 ore. Time Weighted Average TWA: 50 mg/m ³ 8 ore. Time Weighted Average STEL: 50 ppm 15 minuti. Short-term during exposure monitoring, account should be taken of relevant biological monitoring values as suggested by the Scientific Committee on Occupational Exposure Limits for Chemicals Agents (SCOEL) STEL: 250 mg/m ³ 15 minuti. Short-term during exposure monitoring, account should be taken of relevant biological monitoring values as suggested by the Scientific Committee on Occupational Exposure Limits for Chemicals Agents (SCOEL) Pelle	TWA: 10 ppm (8 Stunden). AGW - exposure factor 4 TWA: 50 mg/m ³ (8 Stunden). AGW - exposure factor 4 TWA: 10 ppm (8 Stunden). MAK TWA: 50 mg/m ³ (8 Stunden). MAK Höhepunkt: 40 ppm Höhepunkt: 200 mg/m ³ Haut	STEL: 50 ppm 15 minutos STEL: 250 mg/m ³ 15 minutos TWA: 10 ppm 8 horas TWA: 50 mg/m ³ 8 horas Pele	huid STEL: 50 ppm 15 minuten STEL: 250 mg/m ³ 15 minuten TWA: 10 ppm 8 uren TWA: 50 mg/m ³ 8 uren	TWA: 10 ppm 8 tunteina TWA: 50 mg/m ³ 8 tunteina STEL: 50 ppm 15 minuutteina STEL: 250 mg/m ³ 15 minuutteina Iho

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Acetophenone			TWA: 10 ppm 8 horas		TWA: 5 ppm 8 tunteina TWA: 25 mg/m ³ 8 tunteina
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Component	Austria	Denmark	Switzerland	Poland	Norway
Isopropylbenzene	Haut MAK-KZGW: 50 ppm 15 Minuten MAK-KZGW: 250 mg/m ³ 15 Minuten MAK-TMW: 10 ppm 8 Stunden MAK-TMW: 50 mg/m ³ 8 Stunden	TWA: 10 ppm 8 timer TWA: 50 mg/m ³ 8 timer STEL: 250 mg/m ³ 15 minutter STEL: 50 ppm 15 minutter Hud	Haut/Peau STEL: 80 ppm 15 Minuten STEL: 400 mg/m ³ 15 Minuten TWA: 20 ppm 8 Stunden TWA: 100 mg/m ³ 8 Stunden	STEL: 250 mg/m ³ 15 minutach TWA: 50 mg/m ³ 8 godzinach	TWA: 50 mg/m ³ 8 timer TWA: 10 ppm 8 timer STEL: 250 mg/m ³ 15 minutter. value from the regulation STEL: 50 ppm 15 minutter. value from the regulation Hud
Acetophenone		TWA: 10 ppm 8 timer TWA: 49 mg/m ³ 8 timer STEL: 20 ppm 15 minutter STEL: 98 mg/m ³ 15 minutter		STEL: 100 mg/m ³ 15 minutach TWA: 50 mg/m ³ 8 godzinach	

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Isopropylbenzene	TWA: 10 ppm TWA: 50 mg/m ³ STEL : 50 ppm STEL : 250 mg/m ³ Skin notation	kože TWA-GVI: 10 ppm 8 satima. during the monitoring of exposure the relevant value of biological monitoring shall be taken into account as suggested by the Scientific Committee for Occupational Exposure Limits to Chemical Agents (SCOEL) TWA-GVI: 50 mg/m ³ 8 satima. during the monitoring of exposure the relevant value of biological monitoring shall be taken into account as suggested by the Scientific Committee for Occupational Exposure Limits to Chemical Agents (SCOEL) STEL-KGVI: 50 ppm 15 minutama. STEL-KGVI: 250 mg/m ³ 15 minutama. during the monitoring of exposure the relevant value of biological monitoring shall be taken into account as suggested by the Scientific Committee for Occupational Exposure Limits to Chemical Agents (SCOEL)	TWA: 10 ppm 8 hr. TWA: 50 mg/m ³ 8 hr. STEL: 50 ppm 15 min STEL: 250 mg/m ³ 15 min Skin	Skin-potential for cutaneous absorption STEL: 50 ppm STEL: 250 mg/m ³ TWA: 10 ppm TWA: 50 mg/m ³	TWA: 100 mg/m ³ 8 hodinách. Potential for cutaneous absorption Ceiling: 250 mg/m ³
Benzenemethanol, .alpha.,.alpha.-dimethyl-	TWA: 0.05 mg/m ³				
Acetophenone	TWA: 5.0 mg/m ³		TWA: 10 ppm 8 hr. TWA: 49 mg/m ³ 8 hr. STEL: 30 ppm 15 min STEL: 147 mg/m ³ 15 min		

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Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Isopropylbenzene	Nahk TWA: 10 ppm 8 tundides. TWA: 50 mg/m ³ 8 tundides. STEL: 50 ppm 15 minutites. STEL: 250 mg/m ³ 15 minutites.	Skin notation TWA: 20 ppm 8 hr TWA: 100 mg/m ³ 8 hr STEL: 50 ppm 15 min STEL: 250 mg/m ³ 15 min	skin - potential for cutaneous absorption STEL: 50 ppm STEL: 250 mg/m ³ TWA: 10 ppm TWA: 50 mg/m ³	STEL: 250 mg/m ³ 15 percekben. CK STEL: 50 ppm 15 percekben. CK TWA: 50 mg/m ³ 8 órában. AK TWA: 10 ppm 8 órában. AK lehetséges borón keresztüli felszívódás	STEL: 50 ppm STEL: 250 mg/m ³ TWA: 10 ppm 8 klukkustundum. TWA: 50 mg/m ³ 8 klukkustundum. Skin notation
Acetophenone				TWA: 50 mg/m ³ 8 órában. AK	TWA: 10 ppm 8 klukkustundum. TWA: 49 mg/m ³ 8 klukkustundum. Ceiling: 20 ppm Ceiling: 98 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Cumene hydroperoxide	TWA: 1 mg/m ³	TWA: 1 mg/m ³ IPRD Oda			
Isopropylbenzene	skin - potential for cutaneous exposure STEL: 50 ppm STEL: 250 mg/m ³ TWA: 10 ppm TWA: 50 mg/m ³	TWA: 50 mg/m ³ IPRD in addition to the indicative occupational exposure limit values, biological monitoring values must be taken into account when monitoring exposure TWA: 10 ppm IPRD in addition to the indicative occupational exposure limit values, biological monitoring values must be taken into account when monitoring exposure Oda STEL: 170 mg/m ³ STEL: 35 ppm	Possibility of significant uptake through the skin TWA: 10 ppm 8 Stunden TWA: 50 mg/m ³ 8 Stunden STEL: 50 ppm 15 Minuten STEL: 250 mg/m ³ 15 Minuten	possibility of significant uptake through the skin TWA: 10 ppm TWA: 50 mg/m ³ STEL: 50 ppm 15 minuti STEL: 250 mg/m ³ 15 minuti	Skin notation TWA: 10 ppm 8 ore TWA: 50 mg/m ³ 8 ore STEL: 50 ppm 15 minute STEL: 250 mg/m ³ 15 minute
Acetophenone	TWA: 5 mg/m ³	TWA: 5 mg/m ³ IPRD Oda			TWA: 20 ppm 8 ore TWA: 100 mg/m ³ 8 ore STEL: 41 ppm 15 minute STEL: 200 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Cumene hydroperoxide	Skin notation MAC: 1 mg/m ³				
Isopropylbenzene	TWA: 50 mg/m ³ 1431 MAC: 150 mg/m ³	Ceiling: 250 mg/m ³ Potential for cutaneous absorption TWA: 20 ppm TWA: 500 mg/m ³	TWA: 10 ppm 8 urah TWA: 50 mg/m ³ 8 urah Koža STEL: 50 ppm 15 minutah STEL: 250 mg/m ³ 15 minutah	Binding STEL: 50 ppm 15 minuter Binding STEL: 250 mg/m ³ 15 minuter TLV: 10 ppm 8 timmar. NGV TLV: 50 mg/m ³ 8 timmar. NGV Hud	Deri TWA: 20 ppm 8 saat TWA: 100 mg/m ³ 8 saat STEL: 50 ppm 15 dakika STEL: 250 mg/m ³ 15 dakika
Acetophenone	Skin notation MAC: 5 mg/m ³				

Biological limit values

List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
Isopropylbenzene				2-Phenyl-2-propanol: 7	2-Phenyl-2-propanol

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				mg/g Creatinine urine end of shift	(after hydrolysis): 10 mg/g Creatinine urine (end of shift)
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Component	Italy	Finland	Denmark	Bulgaria	Romania
Isopropylbenzene				2-Phenol-2 propanol: 7 mg/g Creatinine urine up to two hours after the end of work shift possible significant absorption through the skin	

Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Isopropylbenzene		Cumene: 7 µg/g Creatinine urine no later than two hours after the end of the shift	2-Phenylpropane: 10.6 mg/L urine end of exposure or work shift		

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Isopropylbenzene 98-82-8 (7-13)				DNEL = 15.4mg/kg bw/day
Acetophenone 98-86-2 (0.5-1.5)				DNEL = 6.3mg/kg bw/day
Peroxide, bis(1-methyl-1-phenylethyl) 80-43-3 (0.46-0.65)				DNEL = 0.8mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Cumene hydroperoxide 80-15-9 (80-85)				DNEL = 6mg/m³
Isopropylbenzene 98-82-8 (7-13)	DNEL = 250mg/m³			DNEL = 100mg/m³
Acetophenone 98-86-2 (0.5-1.5)				DNEL = 22mg/m³
Peroxide, bis(1-methyl-1-phenylethyl) 80-43-3 (0.46-0.65)				DNEL = 5.6mg/m³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Cumene hydroperoxide 80-15-9 (80-85)	PNEC = 0.0031mg/L	PNEC = 0.023mg/kg sediment dw	PNEC = 0.031mg/L	PNEC = 0.35mg/L	PNEC = 0.0029mg/kg soil dw
Isopropylbenzene	PNEC = 0.035mg/L	PNEC = 3.22mg/kg	PNEC = 0.012mg/L	PNEC = 200mg/L	PNEC =

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98-82-8 (7-13)		sediment dw			0.624mg/kg soil dw
Acetophenone 98-86-2 (0.5-1.5)	PNEC = 0.0864mg/L	PNEC = 0.178mg/kg sediment dw	PNEC = 0.864mg/L	PNEC = 10mg/L	PNEC = 0.155mg/kg soil dw
Peroxide, bis(1-methyl-1-phenylethyl) 80-43-3 (0.46-0.65)	PNEC = 2.34µg/L	PNEC = 2.24mg/kg sediment dw		PNEC = 100mg/L	PNEC = 0.447mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Cumene hydroperoxide 80-15-9 (80-85)	PNEC = 0.00031mg/L	PNEC = 0.0023mg/kg sediment dw			
Isopropylbenzene 98-82-8 (7-13)	PNEC = 0.0035mg/L	PNEC = 0.322mg/kg sediment dw			
Acetophenone 98-86-2 (0.5-1.5)	PNEC = 0.00864mg/L	PNEC = 0.0178mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber Neoprene Natural rubber PVC	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Organic gases and vapours filter Type A Brown conforming to EN14387

Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN

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When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls Prevent product from entering drains. Do not allow material to contaminate ground water system.

Section 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	No information available	
Odor	pungent	
Odor Threshold	No data available	
Melting Point/Range	-30 °C / -22 °F	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	Combustible liquid	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	62 °C / 143 °F	Method - No information available
Autoignition Temperature	380 °C / 716 °F	
Decomposition Temperature	No data available	
Self-Accelerating Decomposition Temperature (SADT)	75°C	
pH	4-7.5	
Viscosity	No data available	
Water Solubility	Miscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Cumene hydroperoxide	1.6	
Isopropylbenzene	3.55	
Acetophenone	1.63 - 1.65	
Peroxide, bis(1-methyl-1-phenylethyl)	5.6	
Vapor Pressure	No data available	
Density / Specific Gravity	1.060	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula	C9 H12 O2
Molecular Weight	152.19
Explosive Properties	explosive air/vapour mixtures possible
Oxidizing Properties	Oxidizer

Section 10: Stability and reactivity

10.1. Reactivity Yes

10.2. Chemical stability Organic peroxide. Hazardous decomposition may occur. Oxidizer: Contact with

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combustible/organic material may cause fire.

10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Temperatures above 40 °C / 104 °F. Excess heat. Do not freeze. Combustible material. Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.

10.5. Incompatible materials

Strong oxidizing agents. Reducing Agent. Acids. Bases. Heavy metals. Strong reducing agents. Combustible material.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO₂).

Section 11: Toxicological information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Category 4
Dermal Category 4
Inhalation Category 3

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Cumene hydroperoxide	LD50 = 382 mg/kg (Rat)	LD50 = 0.126 mL/kg (Rabbit)	LC50 = 220 ppm (Rat) 4 h
Isopropylbenzene	1400 mg/kg (Rat) 2700 mg/kg (Rat)	LD50 = 12300 µL/kg (Rabbit)	LC50 > 3577 ppm (Rat) 6 h
Benzenemethanol, .alpha.,.alpha.-dimethyl-	LD50 = 1300 mg/kg (Rat)	LD50 = 1 mL/kg (Rabbit)	-
Acetophenone	900 mg/kg (Rat) 815 mg/kg (Rat)	3300 mg/kg (Rat)	LC50 > 2.130 mg/L (Rat) 8 h
Peroxide, bis(1-methyl-1-phenylethyl)	LD50 = 4100 mg/kg (Rat)	LD50 > 2000 mg/kg (Rat)	-

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;
Respiratory No data available
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	EU	UK	Germany	IARC
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Isopropylbenzene	Carc Cat. 1B			Group 2B
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(g) reproductive toxicity; Category 1A

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 2

Target Organs Respiratory system, Eyes, Skin, Gastrointestinal tract (GI), Kidney.

(j) aspiration hazard; Category 1

Other Adverse Effects Tumorigenic effects have been reported in experimental animals.

Symptoms / effects, both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

Section 12: Ecological information

12.1. Toxicity

Ecotoxicity effects

The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Cumene hydroperoxide	LC50: = 3.9 mg/L, 96h static (Oncorhynchus mykiss)		
Isopropylbenzene	LC50: = 5.1 mg/L, 96h semi-static (Poecilia reticulata) LC50: = 2.7 mg/L, 96h semi-static (Oncorhynchus mykiss) LC50: 6.04 - 6.61 mg/L, 96h flow-through (Pimephales promelas) LC50: = 4.8 mg/L, 96h flow-through (Oncorhynchus mykiss)	EC50: = 0.6 mg/L, 48h (Daphnia magna) EC50: 7.9 - 14.1 mg/L, 48h Static (Daphnia magna)	EC50: = 2.6 mg/L, 72h (Pseudokirchneriella subcapitata)
Acetophenone	Brachydanio rerio: LC50 = 155 mg/L 96h	EC50 = 162 mg/L 48h	
Peroxide, bis(1-methyl-1-phenylethyl)	LC50: = 15.6 mg/L, 96h (Pimephales promelas) LC50: 80.51 - 146.07 mg/L, 96h semi-static (Poecilia reticulata)		

Component	Microtox	M-Factor
Isopropylbenzene	EC50 = 0.89 mg/L 5 min EC50 = 1.10 mg/L 15 min	

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	EC50 = 1.48 mg/L 30 min EC50 = 172 mg/L 24 h	
Acetophenone	EC50 = 15.5 mg/L 15 min	

12.2. Persistence and degradability

Persistence

Degradation in sewage treatment plant

Not readily biodegradable

Miscible with water, Persistence is unlikely, based on information available.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Cumene hydroperoxide	1.6	35.5 dimensionless
Isopropylbenzene	3.55	35.5 dimensionless
Acetophenone	1.63 - 1.65	No data available
Peroxide, bis(1-methyl-1-phenylethyl)	5.6	137 - 1470 dimensionless 181 - 667 dimensionless

12.4. Mobility in soil

The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment

No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

Section 13: Disposal considerations

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

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Section 14: Transport information

IMDG/IMO

14.1. UN number	UN3109
14.2. UN proper shipping name	ORGANIC PEROXIDE TYPE F, LIQUID
Technical Shipping Name	CUMYL HYDROPEROXIDE
14.3. Transport hazard class(es)	5.2
14.4. Packing group	

ADR

14.1. UN number	UN3109
14.2. UN proper shipping name	ORGANIC PEROXIDE TYPE F, LIQUID
Technical Shipping Name	CUMYL HYDROPEROXIDE
14.3. Transport hazard class(es)	5.2
14.4. Packing group	

IATA

14.1. UN number	UN3109
14.2. UN proper shipping name	ORGANIC PEROXIDE TYPE F, LIQUID
Technical Shipping Name	CUMYL HYDROPEROXIDE
14.3. Transport hazard class(es)	5.2
14.4. Packing group	

14.5. Environmental hazards	Dangerous for the environment Product is a marine pollutant according to the criteria set by IMDG/IMO
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14.6. Special precautions for user	No special precautions required.
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14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods
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Section 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Cumene hydroperoxide	80-15-9	201-254-7	-	-	X	X	KE-24814	X	X
Isopropylbenzene	98-82-8	202-704-5	-	-	X	X	KE-23957	X	X
Benzenemethanol, .alpha.,.alpha.-dimethyl-	617-94-7	210-539-5	-	-	X	X	KE-11212	X	X
Acetophenone	98-86-2	202-708-7	-	-	X	X	KE-28355	X	X
Peroxide, bis(1-methyl-1-phenylethyl)	80-43-3	201-279-3	-	-	X	X	KE-03299	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Cumene hydroperoxide	80-15-9	X	ACTIVE	X	-	X	X	X
Isopropylbenzene	98-82-8	X	ACTIVE	X	-	X	X	X
Benzenemethanol,	617-94-7	X	ACTIVE	X	-	X	X	X

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.alpha.,.alpha.-dimethyl- Acetophenone	98-86-2	X	ACTIVE	X	-	X	X	X
Peroxide, bis(1-methyl-1-phenylethyl)	80-43-3	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Cumene hydroperoxide	80-15-9	-	Use restricted. See entry 75. (see link for restriction details)	-
Isopropylbenzene	98-82-8	-	Use restricted. See entry 28. (see link for restriction details) Use restricted. See entry 75. (see link for restriction details)	-
Benzenemethanol, .alpha.,.alpha.-dimethyl- Acetophenone	617-94-7	-	-	-
Peroxide, bis(1-methyl-1-phenylethyl)	98-86-2	-	Use restricted. See entry 75. (see link for restriction details)	-
	80-43-3	-	Use restricted. See entry 75. (see link for restriction details) Use restricted. See entry 30. (see link for restriction details)	SVHC candidate list - 201-279-3 - Toxic for reproduction, Article 57c

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/candidate-list-table>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Cumene hydroperoxide	80-15-9	Not applicable	Not applicable
Isopropylbenzene	98-82-8	Not applicable	Not applicable
Benzenemethanol, .alpha.,.alpha.-dimethyl- Acetophenone	617-94-7	Not applicable	Not applicable
Peroxide, bis(1-methyl-1-phenylethyl)	98-86-2	Not applicable	Not applicable
	80-43-3	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

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Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Cumene hydroperoxide	WGK2	
Isopropylbenzene	WGK3	
Benzenemethanol, .alpha.,.alpha.-dimethyl-	WGK1	
Acetophenone	WGK1	
Peroxide, bis(1-methyl-1-phenylethyl)	WGK3	

Component	France - INRS (Tables of occupational diseases)
Isopropylbenzene	Tableaux des maladies professionnelles (TMP) - RG 84
Acetophenone	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Isopropylbenzene 98-82-8 (7-13)	Prohibited and Restricted Substances	Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

Section 16: Other information

Full text of H-Statements referred to under sections 2 and 3

H242 - Heating may cause a fire

H302 - Harmful if swallowed

H304 - May be fatal if swallowed and enters airways

H312 - Harmful in contact with skin

H331 - Toxic if inhaled

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

H350 - May cause cancer

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H360D - May damage the unborn child
H373 - May cause damage to organs through prolonged or repeated exposure
H411 - Toxic to aquatic life with long lasting effects
H226 - Flammable liquid and vapor
H315 - Causes skin irritation
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

First aid for chemical exposure, including the use of eye wash and safety showers.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

Chemical incident response training.

Creation Date 02-Feb-2010

Revision Date 29-Oct-2024

Revision Summary SDS sections updated.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and
Preparations).**

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Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet